1. Databases are structured electronic systems used to store and organise data (information) whilst allowing users to access the data, update, manage or modify it easily through the structured digital filing system. The data in the "structured filing system" can be presented in any type of form, including words, numbers, images, videos and files. This can be accomplished through the use of a software called database management systems (DBMS) to store, retrieve and edit data. Thousands of professionals individuals that have knowledge on how to handle data within the working environment can then be safely and simultaneously add, search, or update information.

2. Data warehouse collects and organises data from multiple (different) databases into one central source. Collected from a point of sales system. Relational databases are then complied, cleaned, sorted out before the stored information flows to the data warehouse, making it more accurate, consistent and ready for reports and dashboards that helps with better decision making

3. Data Warehouse & & Data lakes share similar functionalities in terms of storing & processing data, But Data lakes store raw, semi structured and unstructed data from sources such as mobile apps.

4 What is a Data Mart? How does it relate to a data warehouse and typically uses it? (IBM)

Subset of a data warehouse focused on a particular line of business, department or subject area. It improves team efficiency, reduce costs and helps with smarter factual business decision-making in enterprises. Individuals (Users) have access to a subset with specific data saving crucial time as they do not have to search the whole entire data warehouse

5. Explain the difference between structured, Semi-Structured, and Unstructured data. Provide one example of each (AWS)

Structured Data - fits neatly into tables e.g numbers short texts and dates, Relational Data organised formatted repository which typically a database (SQL) storing data in a table (rows + columns) e.g SQL

Unstructured Data - does not fit into table due to nature or size of the file (data). e.g WORD DOC TEXT, MEDIA LOG
Not organised. Non existent predefined manner

Semi-structured Data - does not reside in a relational database but has some organisational properties that make it easier to analyze. e.g XML data
XML used to store and transport data by using custom tags to define a hierarchial structure

6. What is Data Modeling? Why is it important when designing a system that works with data? Describe atleast two types of data models. (AWS)

Process of creating a visual representation or blueprint defining the information collection and management systems of any organization. The BLUEPRINT (Visual representation) / Data model thus helps stakeholders (Engineers, data analysts, scientists, and Electronic engineers) to create a unified view of the organisation's data. The model illustrates what data the business collects, the relationship between different datasets, and methods that will be used to store the & analyze the data. This is important due to large amount of raw data organisations collect. The right applicable structured method used to collect that data, store & process is crucial to guide the business for actionable insights in order to make it profitable (maximising)

Example: Conceptual data model - Overview - provides a big picture view of that data.

- What data system contains
- Data attributes and conditions or constraints on the data
- What business rules the data relates to
- How the data is best organised
- Security and data integrity requirements

Logical Data Model - Maps the conceptual data classes to technical data structures. Provides more detail about the data concepts & complex data relationships identified within the conceptual data.

7. What is Data mining? How is it used in business? Provide two real-world examples of data mining in action. (Aws)

Computer assisted technique used in analytics to process and explore large data sets. Tools & methods can help discover hidden patterns & relationships within the data. Transforms raw data into practical knowledge. Business use it to solve problems, analyze future events that impact the business that require decision making prior to the time which ultimately increases their profit margins

Example: (TELECOM) - Telkom-Company could analyze bandwidth usage patterns & provide customised services upgrades.

(EDUCATION) - Higher Education - Data mining algorithms to build a customized tailored syllabus for students within the rural areas by testing the students and using the applicable methods to collect, process store, analyze the data.

8. What is a Database Management System (DBMS)

software technology managing information within your database. Allows users to access electronically stored data collection which one can read, write, delete and update the data.

MySQL - open source relational database. Organizes data into structured tables with rows & columns
Ideal for web applications and e-commerce platforms

PostgreSQL - advanced open source object relational databases (DBMS) designed for complex data workloads. Ideal for GIS as it can handle complex queries and specialised data types
(Geographical Information System)

Oracle Database - Commercial Multi-model RDBMS including features like robust security & automated tuning through machine learning.
Ideal for mission-critical enterprise applications such as banking systems

9. What is ETL (Extract, Transform, Load)?

3 step data integration process that retrieves raw data from various sources, cleans and structures it to meet business needs, and loads it into a centralised destination like a data warehouse

1. Extract - Raw data copied/read from various sources then stored in a temporary place called "staging area" to avoid slowing down the production source systems.

2. Transform - Raw data processed to fit the target system's requirements. This is the business logic phase:
   - Filtering & Cleaning
   - Standardization
   - Aggregation
   - Joining.

3. Load - Cleaned & Transformed data is loaded into the final destination, typically datawarehouse, data lakes, or databases. This data is now structured + ready for analytical queries

10. Common data storage formats used in analytics (CSV, JSON, Parquet, Avro)? Describe each state

1. CSV (Comma-Separated Values) - A plain text format where each line represents a row and columns are separated by commas. Small datasets (< 1m rows) Used sharing data with non technical users. (EXCEL, CSV readers). Universal! But slow & inefficient.

2. JSON (Javascript Object Notation). lightweight, text based, nested format structured with key-value pairs and arrays. Used for API & WEB SERVICES for excellent data interchanges.
   Ideal for when data is hierarchical (nested) or has varying schemas

3. Parquet

3

11. Artificial Intelligence is the science of creating computer systems capable of performing a task that typically requires human intelligence, such as reasoning, learning, problem solving & understanding language. Machines simulate cognitive functions to solve specific problems or mimic human like decision making.
IBM

Narrow AI (Weak AI) - only type of AI that exists today. Any other form of AI theoretical.

Trained to perform a single or narrow task, often faster than humans.

SIRI & Amazon Alexa are great examples

Performs with its defined task & targets single subset of cognitive abilities and advances them to a spectrum

General AI (AGI) (Strong AI) - theoretical learning concept model. Uses previous learnings + skills to accomplish new tasks without needing human beings to train the underlying models.

12 What is Machine Learning (ML)? How does it differ from traditional programming? Describe the three main types of Machine learning:
IBM
ML is a computer science, data science and artificial intelligence (AI) subset that enables systems to learn and improve from data without additional programming interventions
ML rely on algorithms and statistical models that deploy task based on data patterns and inferences. ML leverage input data to predict outputs, continuously updating outputs as new data becomes available.

Supervised Machine learning - type of ML where the model is trained on the labeled dataset (i.e. target or outcome variable is known)

Data scientists model tornado

Input → Date, location, temperature, wind flow, patterns → Output → Actual tornado output recorded

Mainly used for risk assessment, image recognition, predictive analysis and fraud detection, and comprises several types of algorithms
- Regression algorithms
- Classification algorithms
- Naive Bayes classifiers
- Neural networks
- Random forest algorithms

Unsupervised learning Algorithms - Apriori, Gaussian Mixture Models (GMM) and principal component analysis (PCA)
Draw inferences from unlabled datasets, facilitating exploratory data analysis and enabling pattern recognition and predictive modeling

Reinforcement Learning - Learning from human feedback (RLHF), type of dynamic programming that trains algorithms using a system of reward and punishment

13 What is deep learning? Relation to Machine learning & Artificial intelligence? What neural networks and How do they work?

Deep learning is a subset of machine learning driven by multilayered neural networks whose design is inspired by the structure of the human brain. It excels at recognizing complex patterns. in unstructured data (images, texts)

Unlike ML embedded with a defined mathematical logic of traditional machine learning algorithms, which might perhaps need a human to intervene, the artifical neural networks of the deep learning models comprise of many interconnected layers of neurons that each perform a mathematical operation. Deep learning still uses machine learning to adjust strength of connections between the individual neurons. Deep learning refers to training models with at least 4 layers.

Neural Networks are computing systems inspired by the brains biological neurons. (layout). It consists of layers of interconnected nodes (neurons) that map inputs to ouputs:

14. What is Natural Language Processing (NLP)?

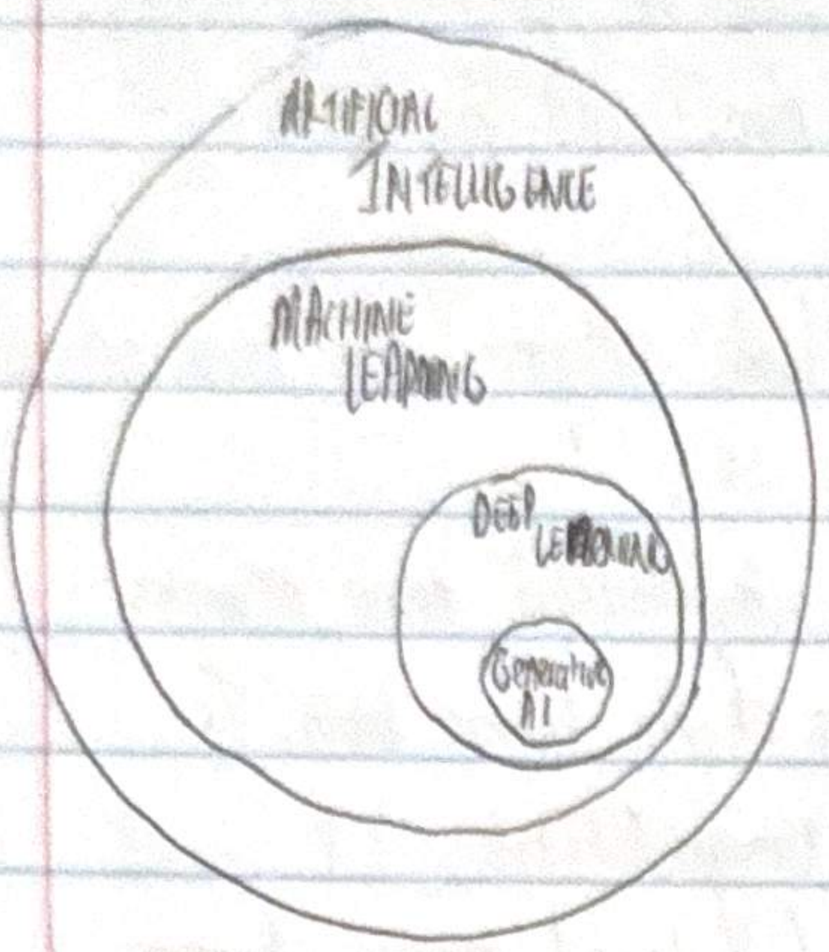
Branch of AI that enable computers understand, interpret, & generate human language, bridging gap between human communication and computer understanding! 3 real world applications are

- Virtual Assistants (Siri, Alexa)
- Autocorrect (Whatsapp embedded autocorrect on IOS)
- Machine Translation (Google Translation)

15. Large Language Model (LLM)? How is it different from standard Machine learning model?

Name two - well known LLMs and explain what they are used for:

Deep learning AI trained on vasts datasets using transformer architecture to understand, summarize, generate, and predict new content, primarily text. It is different to standard ML models due to Scale and Scope: as standard ML usually specialise in narrow tasks! Two well known LLMS - Chat GPT-4 (Open AI) & BERT (Google)



16. Generative AI is a type of AI create new content and ideas, including conversations, stories, images, videos and music. It uses what it knows to solve the next new problem or "next generative idea". At high level it trains large models on vast amounts of data to understand the underlaying structure and patterns of that data. Tools using generative AI

- ChatGPT - conversational AI assistant using Large language Models.
- GitHub Copolito AI pair programmer able to autocomplete code snippets, suggest entire functions & translate natural comments into code.
- Claude (by Anthropic) - AI known for handling large documents (data)

17. What is a training dataset?

Initial, foundational set of data (labeled examples, text, images, figures) used to teach an AI model to recognize patterns, make predictions or generate content. It is important to consider the quality of data as it emcompasses factors less emphasized in traditional data quality dimensions such as representativeness, bias, label accuracy, noise which can ultimately affect how the model works & behaviour. AI models trained on flawed, biased or incomplete data will produce unreliable outputs regardless of how sophisticated architectures might be.

18

18 What is Computer vision? How do machines 'see' and interpret images? Provide two real-world use cases of computer vision? AZURE

Field of AI that enables machines to identify, process and interpret visual data (images and videos) similarly to humans

It uses deep learning and neural networks to recognize objects people and patterns with high degrees of accuracy by replicating human sight and cognitive ability to interpret visual data. Two real world uses include facial recognition, defect detection and self driving vehicles.

19. What is an API (Application Programming Interface)? Explain what it is using a simple real-world analogy, Why are APIs important in modern software & data systems?

Mechanisms (protocols) enabling two softwares components to communicate with each other.

Restuarant Analogy

The Customer (Client/App) : User wants to order a meal

The Kitchen (Server/Database) : Part of the system that prepares your order!

Waiter (API) : Intermediary → Takes your order - request & eventually places it & brings the customers food (response)

The Menu (Documentation) : API documentation tells developer whata data can be requested.

APIs-Important because - Faster Development:
- System Integration:
- Scalability and Flexibility:
- Security
- Innovation:
- Ease of maintenance.

20. API key? Purpose? How it is stored & protected? What could happen if exposed publicly.

IBM

21. An application programming interface (API) that conforms to the design principles of the representational state transfer (REST) architectural style, style used to connect distributed hypermedia systems. Sometimes referred to as RESTful APIs or Restful web APIs.

A REST API would use a Standard HTTP verbs to perform actions on resources.

- GET request to retrieve a record
- POST request to create a new record
- PUT request replace or updates a record
- DELETE request deletes one

} All HTTP methods can be used in API calls

AWS

22. Open data interchange format both human & machine readable, independent of any programming language and is a common API output in a wide variety of applications.

JSON represents data in two ways:

Object: collection of name value (or key value) pairs. Object defined within left ({) and right ({) braces. Each name-value pair begins with the name, followed by a colon, followed by the value. Name-value pairs are comma separated

Array: an ordered collection of values. An array is defined within left ([) and right (]) brackets. Items in the array are comma separated.

JSON example - contain array of objects in which objects represent different films in a streaming library.

1. var films = [{"FilmID": "catmanbegins", "Image": "https://m.media-amazon.com/images/catmanbegins.jpg"}
2. { "FilmID": "cabdriver", "Image": "https://m.media-amazon.com/images/cabdriver.jpg" }

⋮

IBM

23. Art of structuring, designing & redefining inputs (prompts) to guide large language models (LLMs) like ChatGPT & Claude toward generating accurate, relevant, and high-quality outputs.

It helps with efficiency when it comes to completing tasks. It will help construct better inputs when you draft a task for an artifical intelligence programme.

① Poor Prompt: "Write a marketing email for our new office" - MARKETING SECTOR

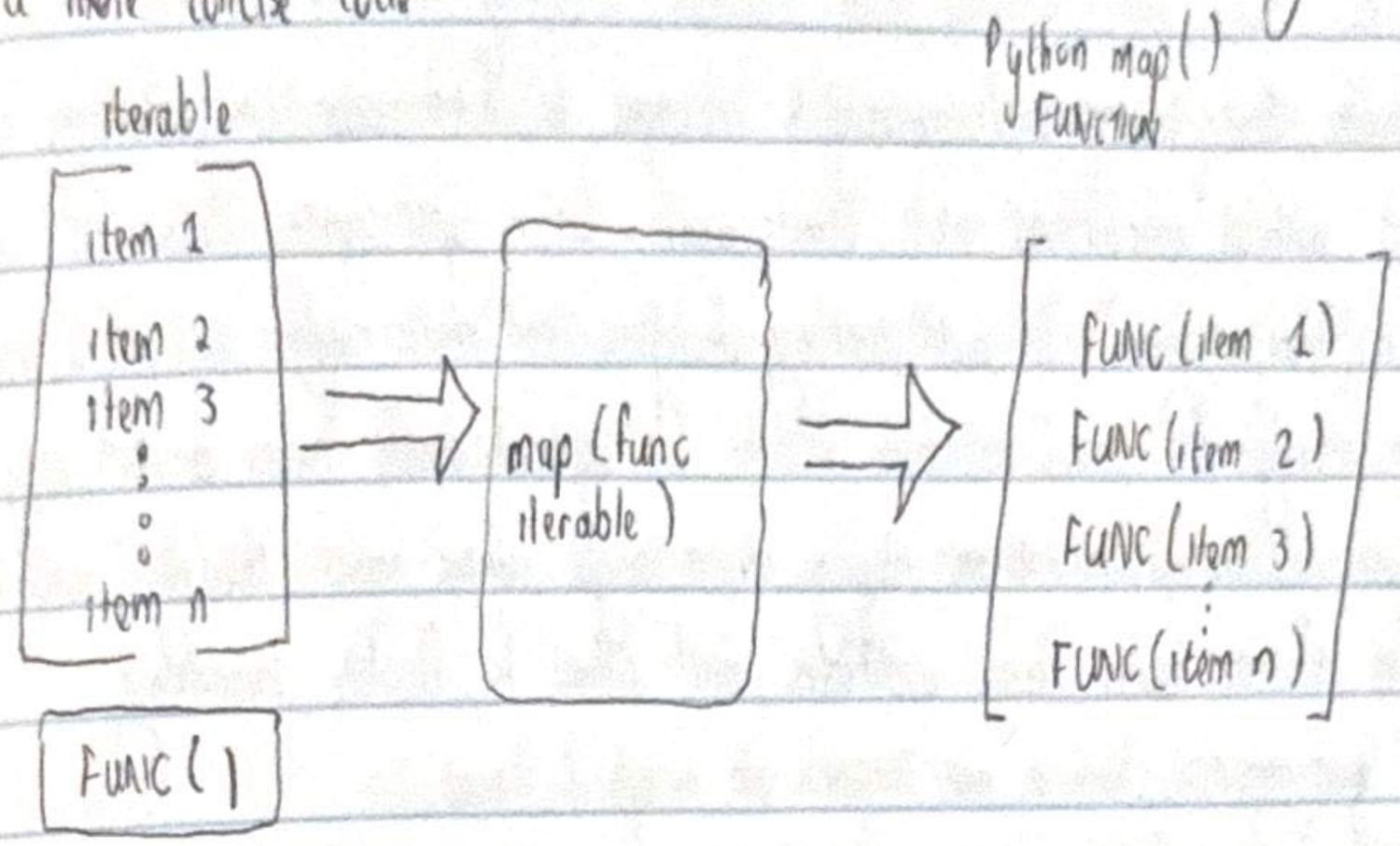
② Poor Prompt: "Make a python function to calculate interest" - CODING - TECH SECTOR

① b) Improved: 150 word email pitch as a professional copy writer. Tone should be energetic and persuasive. Give it a dark roast high caffeine theme.

② b) Improved: The function should include parameters for principal, annual interest rate, and time in years. Returns the total amount which may never be equal to zero

YOUTUBE

24. MAP() function is a higher order function that applies a given transformation to every item in an iterable (like a list or array) and returns a new iterable containing the results. Used for efficient data transformation, allowing developers to update, convert, or process data without writing manual [For] loops, resulting in a more concise code



Python Example

Using map() to square a list of numbers.

```python
numbers = [1, 2, 3, 4]
# Apply lambda x: x**2 to each element in numbers
squared = map(lambda x: x**2, numbers)
print(list(squared)) # Output: [1, 4, 9, 16]
```

JavaScript Example

Using map() to double an array of numbers

```javascript
const numbers = [1, 2, 3, 4];
// Apply arrow function to each element in numbers
const doubled = numbers.map(x => x*2);
console.log(doubled); // Output: [2, 4, 6, 8]
```

25. Cloud computing is the on-demand delivery of IT resources (servers, storage, databases, software) over the internet with pay-as-you-go pricing allowing users to access technology services without managing physical infrastructure

- IaaS (Infrastructure as a service) virtualised computing services over the internet, virtual machines, storage and networking. Eg AWS EC2
- PaaS (Platform as a service) Delivers cloud based environment for developing, testing and deploying applications without managing underlying infrastructure. Eg Google App Engine

SaaS (Software as a service) - ready to use software over the internet, via subscription. The provider manages everything including maintenance, updates and security
1.g Salesforce

26 Version Control System (VCS) software that tracks and manages changes to files over time, allowing multiple people to collaborate on a project without overwriting each other's work. Git is open-source distributed version control system that acts as a "time machine" for code, enabling developers and data analysts to record, compare, and revert changes to projects securely. The main contributor of the file upload would have a full copy of the project history on their local machine, rather than relying on a single central server. Widely used due to the fact that it is fast, efficient with large codebases, and allows for flexible branching workflows "Create a Branch" just means start a new branch on a specific task
"Make Changes & Commit" Edit files and save them (e.g git commit - m "add new data analysis")

27 Jupyter Notebook is an open source, web-based interactive application used to create and share documents that contain live code, equations, visualizations, and narrative text. Functions as digital Notebook where computational work organised into cells that can run individually, providing a flexible environment for both coding and documentation. The name Jupyter is a combination of three original core languages: Julia, Python and Text (R)
Ju, pyt, R (r)
The popular use amongst data scientists & analysts is because it supports the iteractive and visual nature of work

28 Python programming language widely used in web applications, software development, data science, and machine learning (ML). Python widely considered leading language for data analytics & artificial intelligence for several key reasons:
Vast Ecosystem of Libraries
Simple and Readable Syntax
Strong community support
Seamless integration
Standard for AI Research